

Correlation

Introduction

In the previous Chapter we have studied the characteristics of only one variable; example, marks, weights, heights, rainfalls, prices, ages, sales, etc. This type of analysis is called univariate analysis. Sometimes we may be interested to find if there is any relationship between the two variables under study. For example, the price of the commodity and its sale, height of a father and height of his son, price and demand, yield and rainfall, height and weight and so on. Thus the association of any two variables is known as correlation.



Karl Pearson

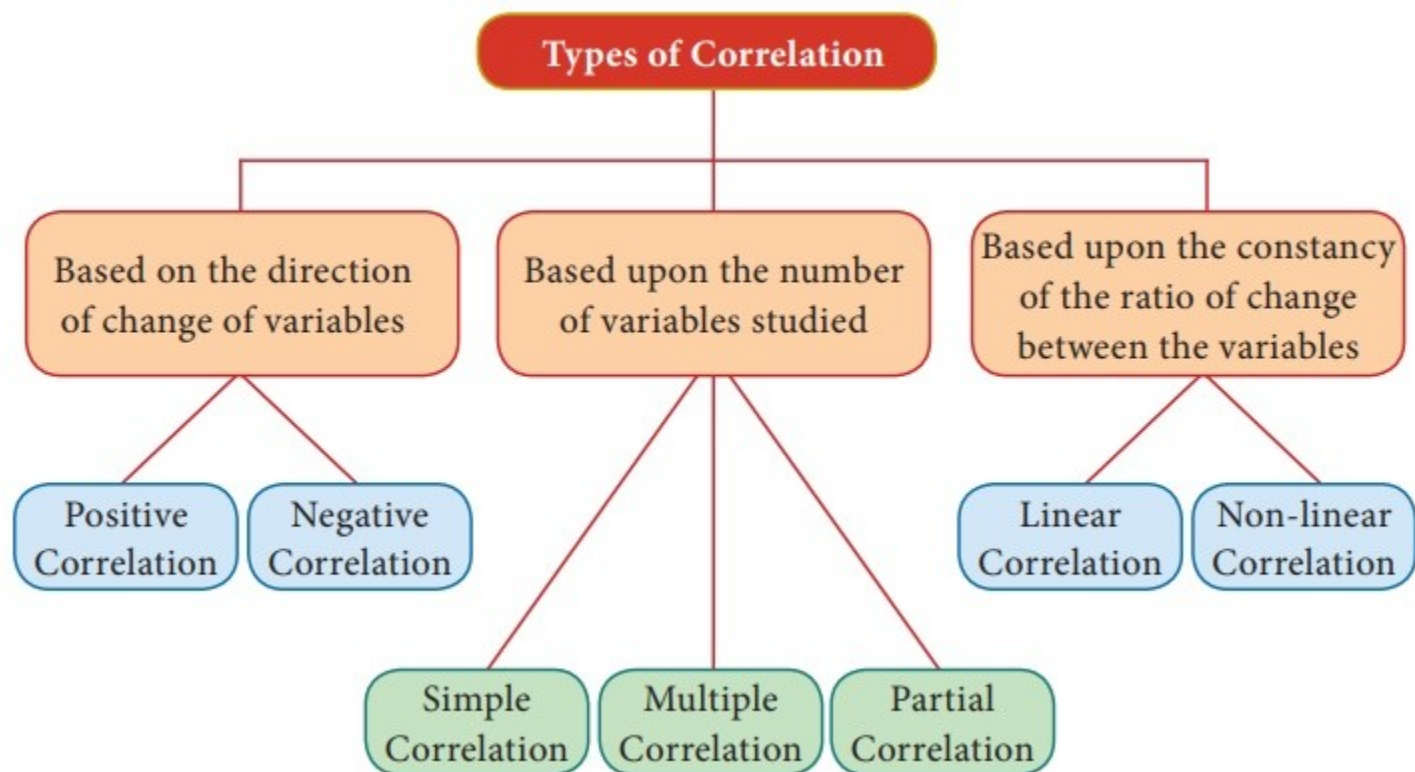
Correlation is the statistical analysis which measures and analyses the degree or extent to which two variables fluctuate with reference to each other.

1. Meaning of Correlation

The term correlation refers to the degree of relationship between two or more variables. If a change in one variable effects a change in the other variable, the variables are said to be correlated.

2. Types of correlation

Correlation is classified in several different ways. Three of the most important ways of classifying correlation are



Correlation is classified into many types, but the important are:

- (i) Positive
- (ii) Negative
- iii) Zero

Positive and negative correlation depends upon the direction of change of the variables.

Positive Correlation

If two variables tend to move together in the same direction that is, an increase in the value of one variable is accompanied by an increase in the value of the other variable; or a decrease in the value of one variable is accompanied by a decrease in the value of the other variable, then the correlation is called positive or direct correlation.

Example

- (i) The heights and weights of individuals
- (ii) Price and Supply
- (iii) Rainfall and Yield of crops
- (iv) The income and expenditure

Negative Correlation

If two variables tend to move together in opposite direction so that an increase or decrease in the values of one variable is accompanied by a decrease or increase in the value of the other variable, then the correlation is called negative or inverse correlation.

Example

- (i) Price and demand

(ii) Repayment period and EMI

(iii) Yield of crops and price

No Correlation

Two variables are said to be uncorrelated if the change in the value of one variable has no connection with the change in the value of the other variable.

For example

We should expect zero correlation (no correlation) between weight of a person and the colour of his hair or the height of a person and the colour of his hair.

Depends on variables correlation are three types:

(i) Simple correlation (ii) Multiple correlation and (iii) partial correlation.

(i) Simple correlation : The degree of relationship between two variables is called simple correlation. Yield of crop per acre and the amount of rainfall, it is a problem of simple correlation.

(ii) Multiple correlation: The degree of relationship among more than two variables is called multiple correlations. Yield of crop per acre and both the amount of rainfall and the amount of fertilizers used, it is a problem of multiple correlation.

(iii) Partial correlation: The degree of relationship between two variables where the other(s) variable(s) is/are held constant. The relationship between Yield of crop per acre and the amount of rainfall where the other variable such as the amount of fertilizers are held constant.

Depends on equations correlation are two types:

(i) Linear correlation (ii) Non linear correlation.

(i) Linear correlation.

If the amount of change in one variable tends to bear a constant ratio to the amount of change in other variable then the correlation is said to be linear. The following example shows the linear correlation.

X: 1 2 3 4 5

Y: 5 10 15 20 25

Here the ratio of change between the two variables is the same. If we plot these points on a graph we get a straight line.

(ii) Non linear correlation.

If the amount of change in one variable does not bear a constant ratio to the amount of change in other variable then the correlation is said to be non linear.

Simple correlation

The correlation between two variables is called simple correlation. The correlation in the case of more than two variables is called multiple correlation.

The following are the mathematical methods of correlation coefficient (r)

(i) Scatter diagram

(ii) Karl Pearson's Coefficient of Correlation

(iii) Rank correlation coefficients

(i) Scatter Diagram

A scatter diagram is the simplest way of the diagrammatic representation of bivariate data. One variable is represented along the X -axis and the other variable is represented along the Y -axis. The pair of points are plotted on the two dimensional graph. The diagram of points so obtained is known as scatter diagram. The direction of flow of points shows the type of correlation that exists between the two given variables. Let $(X_1, Y_1), (X_2, Y_2) \dots (X_N, Y_N)$ be the N pairs of observation of the variables X and Y . If we plot the values of X along x - axis and the corresponding values of Y along y -axis, the diagram so obtained is called a scatter diagram. It gives us an idea of relationship between X and Y . The type of scatter diagram under a simple linear correlation is given below.

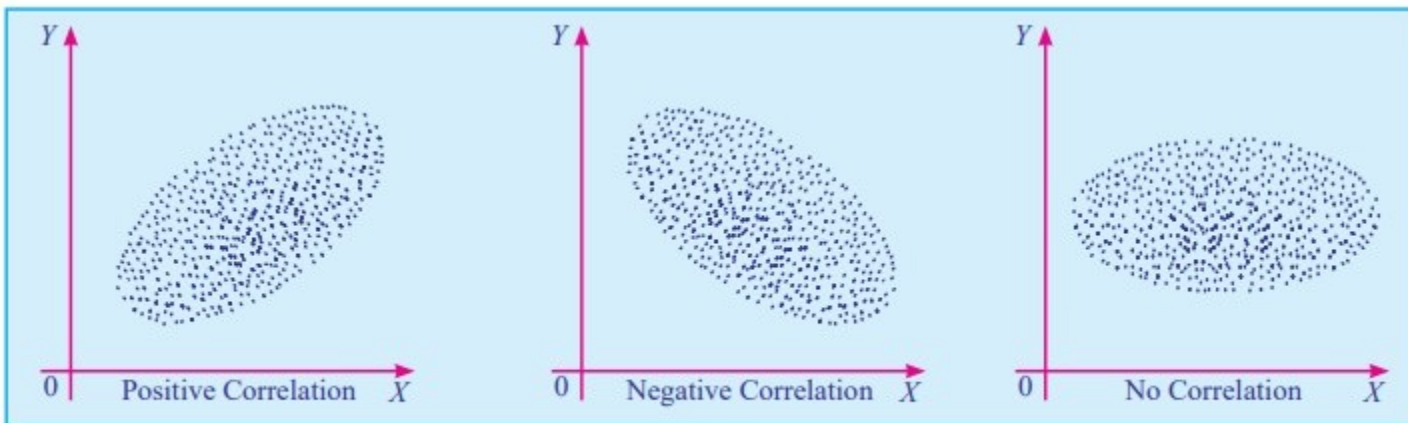
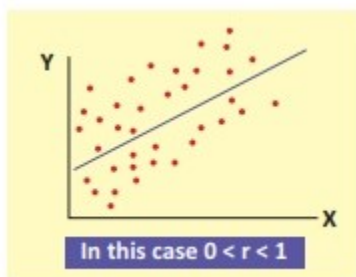


Fig 9.1

- (i) If the plotted points show an upward trend, the correlation will be positive.
- (ii) If the plotted points show a downward trend, the correlation will be negative.
- (iii) If the plotted points show no trend the variables are said to be uncorrelated

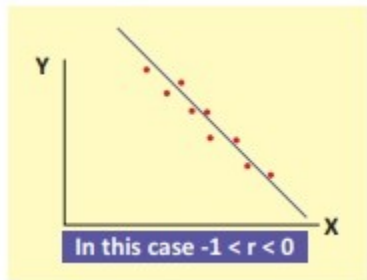
1) Positive correlation

If the plotted points in the plane form a band and they show the rising trend from the lower left hand corner to the upper right hand corner, the two variables are positively correlated.



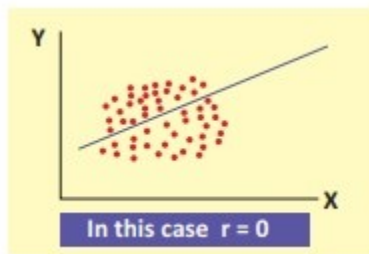
2) Negative correlation

If the plotted points in the plane form a band and they show the falling trend from the upper left hand corner to the lower right hand corner, the two variables are negatively correlated.



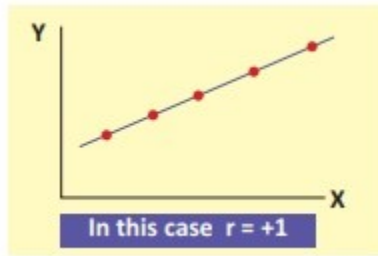
3) Uncorrelated

If the plotted points spread over in the plane then the two variables are uncorrelated.



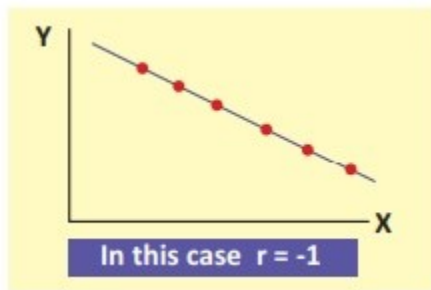
4) Perfect positive correlation

If all the plotted points lie on a straight line from lower left hand corner to the upper right hand corner then the two variables have perfect positive correlation.



5) Perfect Negative correlation

If all the plotted points lie on a straight line falling from upper left hand corner to lower right hand corner, the two variables have perfect negative correlation.



Merits and Demerits of scatter diagram

Merits

It is a simple and non-mathematical method of studying correlation between the variables.

It is not influenced by the extreme items

It is the first step in investigating the relationship between two variables.

It gives a rough idea at a glance whether there is a positive correlation, negative correlation or uncorrelated.

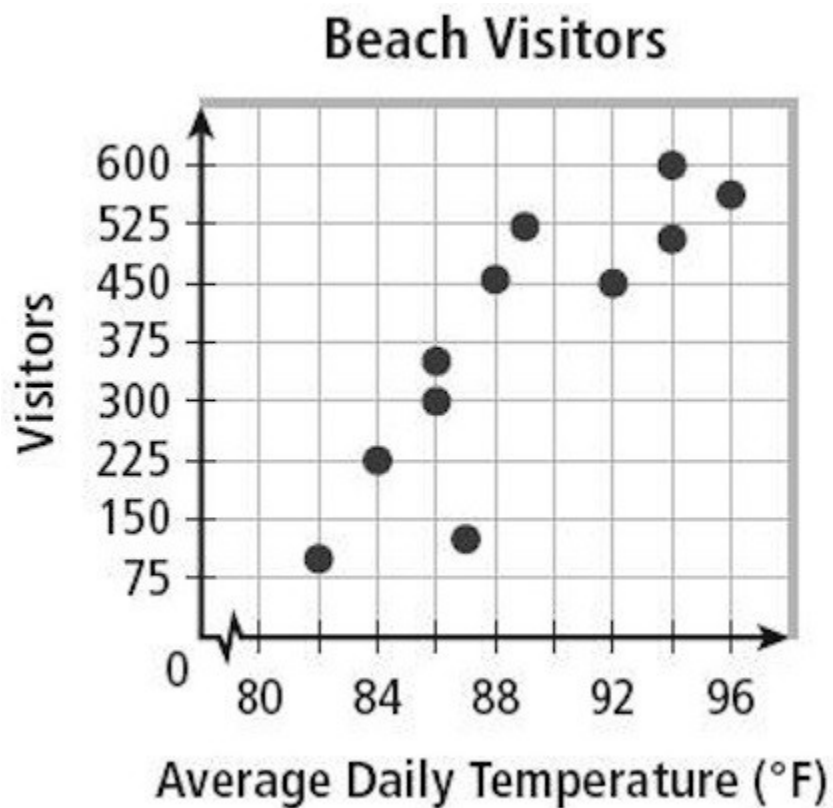
Demerits

We get an idea about the direction of correlation but we cannot establish the exact strength of correlation between the variables.

No mathematical formula is involved.

The given information find correlation coefficient with scatter diagram method.

Daily temp(F)	82	84	94	87	88	94	92	86	86	96	87
Visitors	80	225	600	525	450	480	450	300	325	575	100



Correlation is positive .If average daily temperature is increase then visitors also increase.

Karl Pearson's Correlation Coefficient

Karl Pearson, a great biometrician and statistician, suggested a mathematical method for measuring the magnitude of linear relationship between two variables say X and Y . Karl Pearson's method is the most widely used method in practice and is known as Pearsonian Coefficient of Correlation. It is denoted by the symbol ' r ' and defined as

$$r = \frac{\text{cov}(X, Y)}{\sigma_x \sigma_y}, \text{ where } \text{cov}(X, Y) = \frac{1}{N} \sum (X - \bar{X})(Y - \bar{Y})$$

$$\sigma_x = \sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2}$$

$$\sigma_y = \sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i - \bar{Y})^2}$$

Hence the formula to compute Karl Pearson Correlation coefficient is

$$r = \frac{\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2} \sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i - \bar{Y})^2}}$$

$$r = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^N (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^N (Y_i - \bar{Y})^2}}$$

The coefficient of correlation can also be calculated from the original set of observations (i.e., without taking deviations from mean) by applying the following formula:

$$r = \frac{\sum XY - \frac{\sum X \sum Y}{N}}{\sqrt{\sum X^2 - \frac{(\sum X)^2}{N}} \sqrt{\sum Y^2 - \frac{(\sum Y)^2}{N}}}$$

$$\frac{N \sum XY - \sum X \sum Y}{\sqrt{N \sum X^2 - (\sum X)^2} \sqrt{N \sum Y^2 - (\sum Y)^2}}$$

Example 1

The following data gives the heights(in inches) of father and his eldest son. Compute the correlation coefficient between the heights of fathers and sons using Karl Pearson's method.

<i>Height of father</i>	65	66	67	67	68	69	70	72
<i>Height of son</i>	67	68	65	68	72	72	69	71

Solution:

Let x denote height of father and y denote height of son. The data is on the ratio scale.

We use Karl Pearson's method.

$$r = \frac{n \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{\sqrt{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} \sqrt{n \sum_{i=1}^n y_i^2 - \left(\sum_{i=1}^n y_i \right)^2}}$$

Calculation

x_i	y_i	x_i^2	y_i^2	$x_i y_i$
65	67	4225	4489	4355
66	68	4356	4624	4488
67	65	4489	4225	4355
67	68	4489	4624	4556
68	72	4624	5184	4896
69	72	4761	5184	4968
70	69	4900	4761	4830
72	71	5184	5041	5112
544	552	37028	38132	37560

$$r = \frac{8 \times 37560 - 544 \times 552}{\sqrt{8 \times 37028 - (544)^2} \sqrt{8 \times 38132 - (552)^2}} = 0.603$$

Heights of father and son are positively correlated. It means that on the average, if fathers are tall then sons will probably be tall and if fathers are short, probably sons may be short.

Example 2

The following are the marks scored by 7 students in two tests in a subject. Calculate coefficient of correlation from the following data and interpret.

Marks in test-1	12	9	8	10	11	13	7
Marks in test-2	14	8	6	9	11	12	3

Solution:

Let x denote marks in test-1 and y denote marks in test-2.

	x_i	y_i	x_i^2	y_i^2	$x_i y_i$
	12	14	144	196	168
	9	8	81	64	72
	8	6	64	36	48
	10	9	100	81	90
	11	11	121	121	121
	1	12	169	144	156
	7	3	49	9	21
Total	70	63	728	651	676

$$r = \frac{n \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{\sqrt{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} \sqrt{n \sum_{i=1}^n y_i^2 - \left(\sum_{i=1}^n y_i \right)^2}}$$

$$\sum_{i=1}^n x_i = 70 \quad \sum_{i=1}^n x_i^2 = 728 \quad \sum_{i=1}^n x_i y_i = 676$$

$$\sum_{i=1}^n y_i = 63 \quad \sum_{i=1}^n y_i^2 = 651 \quad n = 7$$

$$\begin{aligned} r &= \frac{7 \times 676 - 70 \times 63}{\sqrt{[7 \times 728 - 70^2]} \times \sqrt{[7 \times 651 - 63^2]}} \\ &= \frac{4732 - 4410}{\sqrt{[5096 - 4900]} \times \sqrt{[7 \times 651 - 3969]}} \\ &= \frac{322}{\sqrt{196} \times \sqrt{588}} = \frac{322}{14 \times 24.25} = \frac{322}{339.5} = 0.95 \end{aligned}$$

There is a high positive correlation between test -1 and test-2. That is those who perform well in test-1 will also perform well in test-2 and those who perform poor in test-1 will perform poor in test- 2.

The students can also verify the results by using shortcut method.

Int

erpretation of Correlation coefficient:

Coefficient of correlation lies between -1 and $+1$. Symbolically, $-1 \leq r \leq +1$

When $r = +1$, then there is perfect positive correlation between the variables.

When $r = -1$, then there is perfect negative correlation between the variables.

When $r = 0$, then there is no relationship between the variables, that is the variables are uncorrelated.

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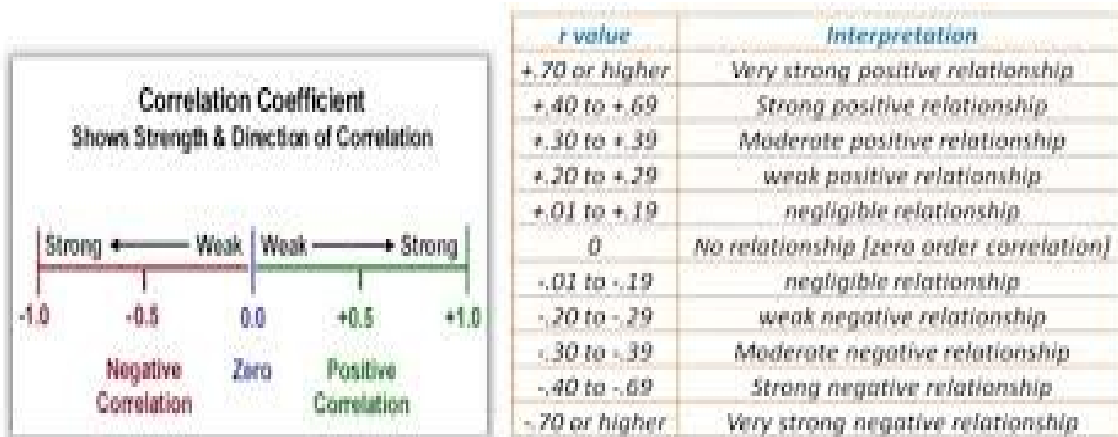
$r = 0.3$, it means there is low degree positive correlation between the variables.

$r=-0.7$, it means there is high degree negative correlation between the variables.

$r=0.02$, it means there is slightly positive correlation between the variables.

$r=0.5$ it means there is moderate positive correlation between the variables.,

Thus, the coefficient of correlation describes the magnitude and direction of correlation.



Methods of computing Correlation Coefficient

(i) When deviations are taken from Mean

Of all the several mathematical methods of measuring correlation, the Karl Pearson's method, popularly known as Pearsonian coefficient of correlation, is most widely used in practice.

$$r = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^N (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^N (Y_i - \bar{Y})^2}} = \frac{\sum xy}{\sqrt{\sum x^2 \sum y^2}}$$

Where $x = (X_i - \bar{X})$ and $y = (Y_i - \bar{Y})$; $i = 1, 2 \dots N$

This method is to be applied only when the deviations of items are taken from actual means.

Steps to solve the problems:

- (i) Find out the mean of the two series that is $\bar{X}$ and $\bar{Y}$.
- (ii) Take deviations of the two series from $\bar{X}$ and $\bar{Y}$ respectively and denoted by x and y .
- (iii) Square the deviations and get the total of the respective squares of deviation of x and y respectively and it is denoted by $\sum x^2$ and $\sum y^2$.
- (iv) Multiply the deviations of x and y and get the total and it is denoted by $\sum xy$.
- (v) Substitute the values of $\sum xy$, $\sum x^2$ and $\sum y^2$ in the above formula.

Example 3

Calculate Karl Pearson's coefficient of correlation from the following data:

X:	6	8	12	15	18	20	24	28	31
Y:	10	12	15	15	18	25	22	26	28

Solution:

X	$x = (X-18)$	x^2	Y	$y = (Y-19)$	y^2	xy
6	-12	144	10	-9	81	108
8	-10	100	12	-7	49	70
12	-6	36	15	-4	16	24
15	-3	9	15	-4	16	12
18	0	0	18	-1	1	0
20	2	4	25	6	36	12
24	6	36	22	3	9	18
28	10	100	26	7	49	70
31	13	169	28	9	81	117
$\sum X = 162$	$\sum x = 0$	$\sum x^2 = 598$	$\sum Y = 171$	$\sum y = 0$	$\sum y^2 = 338$	$\sum xy = 431$

Table 9.1

$$N=9, \bar{X} = \frac{\sum X}{N} = \frac{162}{9} = 18, \bar{Y} = \frac{\sum Y}{N} = \frac{171}{9} = 19$$

$$r = \frac{\sum xy}{\sqrt{\sum x^2 \sum y^2}}$$

where $x=(X-\bar{X})$ and $y=(Y-\bar{Y})$

$$\sum xy = 431, \sum x^2 = 598, \sum y^2 = 338$$

$$r = \frac{431}{\sqrt{598 \times 338}} = \frac{431}{449.582} = +0.959$$

(ii) When actual values are taken (without deviation)

when the values of X and Y are considerably small in magnitude the following formula can be used

$$r = \frac{N\sum XY - (\sum X)(\sum Y)}{\sqrt{N\sum X^2 - (\sum X)^2} \times \sqrt{N\sum Y^2 - (\sum Y)^2}}$$

Example 4

Calculate coefficient of correlation from the following data

X	12	9	8	10	11	13	7
Y	14	8	6	9	11	12	3

Solution:

In both the series items are in small number. Therefore correlation coefficient can also be calculated without taking deviations from actual means or assumed mean.

X	Y	X ²	Y ²	XY
12	14	144	196	168
9	8	81	64	72
8	6	64	36	48
10	9	100	81	90
11	11	121	121	121
13	12	169	144	156
7	3	49	9	21
$\sum X = 70 \quad \sum Y = 63 \quad \sum X^2 = 728 \quad \sum Y^2 = 651 \quad \sum XY = 676$				

Table 9.2

$$\begin{aligned} r &= \frac{N\sum XY - (\sum X)(\sum Y)}{\sqrt{N\sum X^2 - (\sum X)^2} \times \sqrt{N\sum Y^2 - (\sum Y)^2}} \\ &= \frac{7(676) - (70)(63)}{\sqrt{7(728) - (70)^2} \times \sqrt{7(651) - (63)^2}} \\ &= \frac{322}{339.48} \\ r &= +0.95 \end{aligned}$$

Example 5

Determine the value of 'r' from the following data:

X	2	4	3	8	6	5	6	7	5	4
Y	15	20	16	6	1 4	12	12	9	11	15

Solution:

Calculation of the value of 'r'

x	y	x^2	y^2	xy
2	15	4	225	30
4	20	16	400	80
3	16	9	256	48
8	6	64	36	48
6	14	36	196	84
5	12	25	144	60
6	12	36	144	72
7	9	49	81	63
5	11	25	121	55
4	15	16	225	60
$\Sigma x = 50$	$\Sigma y = 130$	$\Sigma x^2 = 280$	$\Sigma y^2 = 1828$	$\Sigma xy = 600$

$$\begin{aligned}\therefore r &= \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{n(\sum x^2) - (\sum x)^2} \cdot \sqrt{n(\sum y^2) - (\sum y)^2}} \\ &= \frac{10(600) - (50)(130)}{\sqrt{10(280) - (50)^2} \cdot \sqrt{10(1828) - (130)^2}} \\ &= -0.78 \quad \text{[Ans.] Problem 1}\end{aligned}$$

Example 6

The following are data on Advertising Expenditure (in Tk Thousand) and Sales (Tk in lakhs) in a company.

Advertising Expenditure	:	18	19	20	21	22	23
Sales	:	17	17	18	19	19	19

Determine the correlation coefficient between them and interpret the result.

Solution:

We have $N = 6$. Calculate $\sum X$, $\sum Y$, $\sum XY$, $\sum Y^2$, $\sum X^2$ as follows:

X	Y	XY	X^2	Y^2
18	17	306	324	289
19	17	323	361	289
20	18	360	400	324
21	19	399	441	361
22	19	418	484	361
23	19	437	529	361
Total :123	109	2243	2539	1985

The correlation coefficient r between the two variables is calculated as follows:

$$r = \frac{N \sum XY - (\sum X)(\sum Y)}{\sqrt{N \sum X^2 - (\sum X)^2} \sqrt{N \sum Y^2 - (\sum Y)^2}}$$

$$r = \frac{6 \times 2243 - 123 \times 109}{\sqrt{6 \times 2539 - (123)^2} \sqrt{6 \times 1985 - (109)^2}}$$

$$= (13458 - 13407) / \{ \sqrt{(15234 - 15129)} \sqrt{(11910 - 11881)} \}$$

$$= 51 / \{ \sqrt{105} \sqrt{29} \} = 51 / (10.247 \times 5.365)$$

$$= 51 / 54.975$$

$$= 0.9277$$

Interpretation

The value of r is 0.92. It shows that there is a high, positive correlation between the two variables 'Advertising Expenditure' and 'Sales'. This provides a basis to consider some functional relationship between them.

Example 6

Calculate the correlation coefficient from the following data

$$N=9, \sum X=45, \sum Y=108, \sum X^2=285, \sum Y^2=1356, \sum XY=597$$

Solution

Example 7

From the following data calculate the correlation coefficient $R_{xy}=120$, $R_{x^2}=90$, $R_{y^2}=640$

Solution:

Given $R_{xy}=120$, $R_{x^2}=90$, $R_{y^2}=640$

$$\begin{aligned} \text{Then} \quad r &= \frac{\sum xy}{\sqrt{\sum x^2 \sum y^2}} = \frac{120}{\sqrt{90(640)}} = \frac{120}{\sqrt{57600}} = \frac{120}{240} = 0.5 \\ \therefore \quad r &= 0.5 \end{aligned}$$

Prove that $-1 \leq r \leq +1$.

Solution: We know,

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

Let
$$\Sigma u = \frac{\sum (x - \bar{x})}{\sqrt{\sum (x - \bar{x})^2}}$$

$$\Rightarrow \Sigma u^2 = \frac{\sum (x - \bar{x})^2}{\sum (x - \bar{x})^2}$$

$$\Rightarrow \Sigma u^2 = 1$$

and
$$\Sigma v = \frac{\sum (y - \bar{y})}{\sqrt{\sum (y - \bar{y})^2}}$$

$$\Rightarrow \Sigma v^2 = \frac{\sum (y - \bar{y})^2}{\sum (y - \bar{y})^2}$$

$$\Rightarrow \Sigma v^2 = 1$$

and
$$\Sigma uv = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

$$= r$$

We know,

$$\Sigma (u \pm v)^2 \geq 0$$

$$\Rightarrow \Sigma (u^2 \pm 2uv + v^2) \geq 0$$

$$\Rightarrow \Sigma u^2 \pm 2 \Sigma uv + \Sigma v^2 \geq 0$$

$$\Rightarrow 1 \pm 2r + 1 \geq 0$$

$$\Rightarrow 2 \pm 2r \geq 0$$

$$\Rightarrow 2(1 \pm r) \geq 0$$

$$\Rightarrow 1 \pm r \geq 0$$

If $1+r \geq 0$, then $r \geq -1$(i)

If $1-r \geq 0$, then $r \leq 1$ (ii)

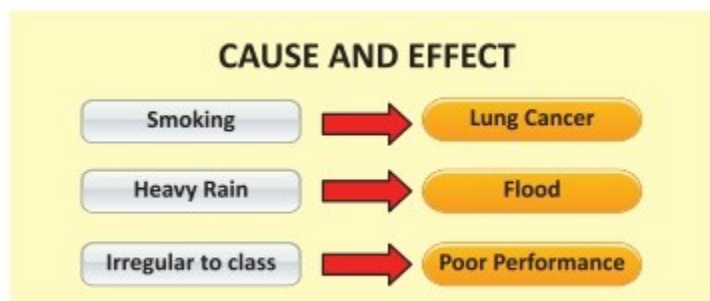
From (i) and (ii), we get

$$-1 \leq r \leq +1. \text{ (Proved)}$$

Limitations of Correlation

Although correlation is a powerful tool, there are some limitations in using it:

1. Outliers (extreme observations) strongly influence the correlation coefficient. If we see outliers in our data, we should be careful about the conclusions we draw from the value of r . The outliers may be dropped before the calculation for meaningful conclusion.
2. Correlation does not imply causal relationship. That a change in one variable causes a change in another.



Probable Error in Coefficient of Correlation & Its Interpretation:

The probable error of the coefficient of correlation helps in interpreting its value. With the help of probable error it is possible to determine the liability of the value of the coefficient so far as it depends on the condition of random sampling.

The formula for finding out the probable error of the Karl Pearson's coefficient of correlation is:

Probable error of coefficient of correlation

$$P.E.r = 0.6745 \frac{1-r^2}{\sqrt{N}}$$

- (i) If the value of r is less than the probable error, there is no evidence of correlation.
- (ii) If the value of r is more than six times of the probable error it is significant correlation.
- (iii) If the probable error is not much and if the coefficient of correlation is .5 or more it is generally considered to be significant.
- (iv) Probable error as a measure for interpreting coefficient of correlation should be used only when the number of pairs of observations is large. If n is small probable error may give misleading conclusions.
- (v) Probable error as a measure for interpreting coefficient of correlation should be used only when a sample study is being made and the sample is unbiased and representative. With the help of sample studies, the correlation in the universe is estimated.

Rank Correlation coefficient:

It is studied when no assumption about the parameters of the population is made. This method is based on ranks. It is useful to study the qualitative

measure of attributes like honesty, color, beauty, intelligence, character, morality etc. Spearman Rank Correlation Coefficient is a non-parametric measure of correlation, using ranks to calculate the correlation. The individuals in the group can be arranged in order and there on, obtaining for each individual a number showing his/her rank in the group. This method was developed by Edward Spearman in 1904. It is defined by

$$R = 1 - \frac{6 \sum D^2}{N(N^2 - 1)} = 1 - \frac{6 \sum D^2}{N^3 - N}$$

Where, R denotes rank correlation coefficient and D refers to the difference of ranks between paired items in two series.

Proof:

Let us suppose that a group of n individuals is arranged in order of merit or proficiency in possession of two characteristics A and B. these ranks in two characteristics will, in general, be different. For example, if we consider the relation between intelligence and beauty, it is not necessary that a beautiful individual is intelligent also. Let (x_i, y_i) ; $i=1, 2, \dots, n$ be the ranks of the i th individual in two characteristics A and B respectively. Pearsonian coefficient of correlation between the ranks x_i 's and y_i 's is called the rank correlation coefficient between A and B for that group of individuals.

Assuming that no two individuals are bracketed equal in either classification, each of the variables X and Y takes the values 1, 2, ..., n

$$x_i = 1, 2, 3, \dots, n$$

$$y_i = 1, 2, 3, \dots, n \quad \sum x_i = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} = \sum y_i$$

$$\bar{x} = \frac{n(n+1)}{2} / n = \frac{(n+1)}{2} = \bar{y}$$

$$\sum x_i^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} = \sum y_i^2$$

We know .correlation coefficient

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} \dots\dots\dots(1)$$

$$\sum (x - \bar{x})^2 = \sum x^2 - (\sum x/n)^2$$

$$= \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)^2}{12} = \sum (y - \bar{y})^2$$

We know

$$d_i = x_i - y_i$$

$$\sum d_i^2 = \sum [(x_i - \bar{x}) - (y_i - \bar{y})]^2$$

$$= \sum (x - \bar{x})^2 - 2 \sum (x - \bar{x})(y - \bar{y}) + \sum (y - \bar{y})^2$$

$$= \frac{n(n+1)^2}{12} - 2 \sum (x - \bar{x})(y - \bar{y}) + \frac{n(n+1)^2}{12} \text{ [put the value]}$$

$$2 \sum (x - \bar{x})(y - \bar{y}) = 2 \frac{n(n+1)^2}{12} - \sum d_i^2$$

$$\sum (x - \bar{x})(y - \bar{y}) = \frac{n(n+1)^2}{12} - \sum d_i^2 / 2$$

Put the value in equation (1)

$$r = \frac{n(n+1)^2}{12} - \frac{\sum d_i^2}{2} \bigg/ \frac{n(n+1)^2}{12}$$

$$R = 1 - \frac{6 \sum di^2}{n(n^2 - 1)} = 1 - \frac{6 \sum di^2}{n^3 - n}$$

Which is the spearman's formula for the rank correlation coefficient.

Problem 1

The coefficient of rank correlation of the marks obtained by 10 students in statistics and accountancy was found to be 80. It was later discovered that the difference in ranks in the two subjects obtained by one of the students was wrongly take as 7 instead of 9. Find the correct coefficient of rank correlation.

Solution: Given $N = 10$

$$R = 0.80$$

$$\therefore R = 1 - \frac{6 \sum D^2}{N^3 - N}$$

$$\Rightarrow 0.80 = 1 - \frac{6 \sum D^2}{10^3 - 10}$$

$$\Rightarrow \frac{6 \sum D^2}{990} = 1 - 0.80$$

$$\Rightarrow \frac{\sum D^2}{165} = 0.20$$

$$\Rightarrow \sum D^2 = 33, \text{ but this is not correct.}$$

$$\therefore \text{Correct } \sum D^2 = 33 - 7^2 + 9^2 = 65$$

$$\frac{6 \sum D^2}{N^3 - N} = 1 - \frac{6 \times 65}{10^3 - 10} = 1 - 0.394 = 0.606$$

$$\therefore R = 1 -$$

[Ans.]

Example 2

Two referees in a flower beauty competition rank the 10 types of flowers as follows:

Referee A	1	6	5	10	3	2	4	9	7	8
Referee B	6	4	9	8	1	2	3	10	5	7

Use the rank correlation coefficient and find out what degree of agreement is between the referees.

Solution:

Rank by 1 st referee R_{1i}	Rank by 2 nd referee R_{2i}	$D_i = R_{1i} - R_{2i}$	D_i^2
1	6	-5	25
6	4	2	4
5	9	-4	16
10	8	2	4
3	1	2	4
2	2	0	0
4	3	1	1
9	10	-1	1
7	5	2	4
8	7	1	1
			$\sum_{i=1}^n D_i^2 = 60$

Here $n = 10$ and $\sum_{i=1}^n D_i^2 = 60$

$$\begin{aligned}\rho &= 1 - \frac{6 \sum_{i=1}^n D_i^2}{n(n^2 - 1)} \\ &= 1 - \frac{6 \times 60}{10(10^2 - 1)} = 1 - \frac{360}{10(99)} = 1 - \frac{360}{990} = 0.636\end{aligned}$$

Interpretation: Degree of agreement between the referees ‘A’ and ‘B’ is 0.636 and they have “strong agreement” in evaluating the competitors.

Repeated ranks

When two or more items have equal values (i.e., a tie) it is difficult to give ranks to them. In such cases the items are given the average of the ranks they would have received. For example, if two individuals are placed in the 8th place, they are given the rank $[8+9] / 2 = 8.5$ each, which is common rank to be assigned and the next will be 10; and if three ranked equal at the 8th place, they are given the rank $[8 + 9 + 10] / 3 = 9$ which is the common rank to be assigned to each; and the next rank will be 11.

In this case, a different formula is used when there is more than one item having the same value.

$$\rho = 1 - 6 \left[\frac{\sum D_i^2 + \frac{1}{12}(m_1^3 - m_1) + \frac{1}{12}(m_2^3 - m_2) + \dots}{n(n^2 - 1)} \right]$$

where m_i is the number of repetitions of i^{th} rank

Example 3

Compute the rank correlation coefficient for the following data of the marks obtained by 8 students in the Commerce and Mathematics.

Marks in Commerce	15	20	28	12	40	60	20	80
Marks in Mathematics	40	30	50	30	20	10	30	60

Solution:

Marks in Commerce (X)	Rank (R_{1i})	Marks in Mathematics (Y)	Rank (R_{2i})	$D_i = R_{1i} - R_{2i}$	D_i^2
15	2	40	6	-4	16
20	3.5	30	4	-0.5	0.25
28	5	50	7	-2	4
12	1	30	4	-3	9
40	6	20	2	4	16
60	7	10	1	6	36
20	3.5	30	4	-0.5	0.25
80	8	60	8	0	0
				Total	$\sum D^2 = 81.5$

$$\rho = 1 - 6 \left[\frac{\sum D_i^2 + \frac{1}{12}(m_1^3 - m_1) + \frac{1}{12}(m_2^3 - m_2) + \dots}{n(n^2 - 1)} \right]$$

Repetitions of ranks

In Commerce (X), 20 is repeated two times corresponding to ranks 3 and 4. Therefore, 3.5 is assigned for rank 2 and 3 with $m_1=2$.

In Mathematics (Y), 30 is repeated three times corresponding to ranks 3, 4 and 5. Therefore, 4 is assigned for ranks 3,4 and 5 with $m_2=3$.

Therefore,

$$\rho = 1 - 6 \left[\frac{81.5 + \frac{1}{12} (2^3 - 2) + \frac{1}{12} (3^3 - 3)}{8 (8^2 - 1)} \right]$$

$$= 1 - 6 \left[\frac{81.5 + 0.5 + 2}{504} \right] = 1 - \frac{504}{504} = 0$$

Interpretation: Marks in Commerce and Mathematics are uncorrelated

Problem 4

A psychologist wanted to compare two methods A and B of teaching. He selected a random sample of 22 students. He grouped them into 11 pairs so that the students in a pair have approximately equal scores on an intelligence test. In each pair one student was taught by method A and the other by method B and examined after the course. The marks obtained by them are tabulated below:

Pair	1	2	3	4	5	6	7	8	9	10	11
A	24	29	19	14	30	19	27	30	20	28	11
B	37	35	16	26	23	27	19	20	16	11	21

Find the rank correlation coefficient between the two sets of scores.

Solution: (a) Calculation of Rank correlation coefficient

Ranking No.	A	Method A (R_1)	B	Method B (R_2)	$(R_1 - R_2)^2 = D^2$
-------------	---	--------------------	---	--------------------	-----------------------

1	2	6	37	11	25
2	4	9	35	10	1
3	2	3.5	16	2.5	1
4	9	2	26	8	36
5	1	10.5	23	7	12.25
6	9	3.5	27	9	30.25
7	4	7	19	4	9
8	3	10.5	20	5	30.25
9	0	5	16	2.5	6.25
10	1	8	11	1	49
11	9	1	21	6	25
	2				
	7				
	3				
	0				
	2				
	0				
	2				
	8				
	1				
	1				
N=11					ΣD²=225

$$R=1-\frac{6\left\{\Sigma D^2+\frac{1}{12}(m^3-m)+\frac{1}{12}(m^3-m)+\frac{1}{12}(m^3-m)\right\}}{N^3-N}$$

$$= 1 - \frac{6 \left\{ 225 + \frac{1}{12}(2^3 - 2) + \frac{1}{12}(2^3 - 2) + \frac{1}{12}(2^3 - 2) \right\}}{11^3 - 11}$$

$$= -0.0295 \text{ [Ans.]}$$

Problem 5

Ten competitors in a music contest are ranked by three Judges in the following order:

First Judge	4	6	9	5	1	3	10	8	7	2
Second Judge	4	8	7	5	6	9	10	3	2	1
Third Judge	7	8	6	1	5	1	9	2	3	4
						0				

Use the rank correlation coefficient to discuss which pair of Judges has the nearest approach to common tastes in music?

Solution: Calculation of Rank correlation coefficient

Judge 1 R_1	Judge 2 R_2	Judge 3 R_3	$(R_1 - R_2)^2$ $= D^2$	$(R_2 - R_3)^2$ $= D^2$	$(R_1 - R_3)^2$ $= D^2$
4	4	7	0	9	9
6	8	8	4	0	4
9	7	6	4	1	9
5	5	1	0	16	16
1	6	5	25	1	16
3	9	10	36	1	49
10	10	9	0	1	1
8	3	2	25	1	36
7	2	3	25	1	16
2	1	4	1	9	4
N = 10			$\Sigma D^2 = 120$	$\Sigma D^2 = 40$	$\Sigma D^2 = 160$

$$R(1 \text{ \& } 2) = 1 - \frac{6 \Sigma D^2}{N^3 - N} = 1 - \frac{6 \times 120}{10^3 - 10} = 1 - 0.73 = 0.27$$

$$R(2 \text{ \& } 3) = 1 - \frac{6 \Sigma D^2}{N^3 - N} = 1 - \frac{6 \times 40}{10^3 - 10} = 1 - 0.24 = 0.76$$

$$R(1 \text{ \& } 3) = 1 - \frac{6 \Sigma D^2}{N^3 - N} = 1 - \frac{6 \times 160}{10^3 - 10} = 1 - 0.97 = 0.03$$

Comment: Since Judge 2 and Judge 3 have the highest positive the rank correlation coefficient, so they have the nearest approach to common tastes in music.

Problem 6

The coefficient of rank correlation between debenture prices and share prices is found 0.143 and the sum of squares of the difference is 48. Find out the value of N (Number of pairs).

Solution: Given $R = 0.143$, $\Sigma D^2 = 48$, $N = ?$

We know,

$$R = \frac{1 - \frac{6 \Sigma D^2}{N^3 - N}}$$

$$\Rightarrow 0.143 = \frac{1 - \frac{6 \times 48}{N^3 - N}}$$

$$\Rightarrow 0.857 = \frac{288}{N^3 - N}$$

$$\Rightarrow N^3 - N = \frac{288}{0.857}$$

$$\Rightarrow N^3 - N = 336$$

$$\Rightarrow N^3 - N - 336 = 0$$

$$\Rightarrow N^3 - 7N^2 + 7N^2 - 49N + 48N - 336 = 0$$

$$\Rightarrow N^2(N - 7) + 7N(N - 7) + 48(N - 7) = 0$$

$$\Rightarrow (N - 7)(N^2 + 7N + 48) = 0$$

$$\Rightarrow (N - 7) = 0$$

$$\Rightarrow N = 7 \quad [\text{Ans.}]$$

What is 'Regression'

Regression is a statistical measure used in finance, investing and other disciplines that attempts to determine the strength of the relationship between one dependent variable (usually denoted by Y) and a series of other changing variables (known as independent variables). Regression helps investment and financial managers to value assets and understand the relationships between variables, such as commodity prices and the stocks of businesses dealing in those commodities.

The two basic types of regression are linear regression and multiple linear regression, although there are non-linear regression methods for more complicated data and analysis. Linear regression uses one independent variable to explain or predict the outcome of the dependent variable Y, while multiple regression uses two or more independent variables to predict the outcome.

Regression can help finance and investment professionals as well as professionals in other businesses. Regression can help predict sales for a company based on weather, previous sales, GDP growth or other

conditions. The capital asset pricing model (CAPM) is an often-used regression model in finance for pricing assets and discovering costs of capital. The general form of each type of regression is:

Linear Regression: $Y = a + bX + u$

Multiple Regression: $Y = a + b_1X_1 + b_2X_2 + b_3X_3 + \dots + b_tX_t + u$

Where:

Y = the variable that you are trying to predict (dependent variable)

X = the variable that you are using to predict Y (independent variable)

a = the intercept

b = the slope

u = the regression residual

Regression takes a group of random variables, thought to be predicting Y, and tries to find a mathematical relationship between them. This relationship is typically in the form of a straight line (linear regression) that best approximates all the individual data points. In multiple regression, the separate variables are differentiated by using numbers with subscript.

Regression Equation

Now that we know how the relative relationship between the two variables is calculated, we can develop a regression equation to forecast or predict the variable we desire. Below is the formula for a simple linear regression. The "y" is the value we are trying to forecast, the "b" is the slope of the regression, the "x" is the value of our independent value, and the "a" represents the y-intercept. The regression equation simply describes the relationship between the dependent variable (y) and the independent variable (x).

$$y = bx +$$

The intercept, or "a", is the value of y (dependent variable) if the value of x (independent variable) is zero. So if there was no change in GDP, your company would still make some sales - this value, when the change in GDP is zero, is the intercept. Take a look at the graph below to see a graphical depiction of a

regression equation. In this graph, there are only five data points represented by the five dots on the graph. Linear regression attempts to estimate a line that best fits the data, and the equation of that line results in the regression equation.

Utility of Regression Analysis

Some of the Utility of Regression Analysis are as follows:

1. Regression analysis helps in establishing a functional relationship between two or more variables.
2. Since most of the problems of economic analysis are based on cause and effect relationship, the regression analysis is a highly valuable tool in economics & business research.
3. Regression analysis is widely used in statistical estimation of demand curves, supply curves, production functions, cost functions, consumption functions etc.

Regression Equation of Y on X

The regression equation of Y on X is expressed as follows:

$$Y_e = a + bX$$

The regression equation of X on Y is expressed as follows:

$$X_c = a + bY$$

In this equation a and b are two unknown constants (fixed numerical values) which determine the position of the line completely. These constants are called the parameters of the line. If the value of either or both of them is changed, another line is determined. The parameter ' a ' determines the level of the fitted line (i.e., the distance of the line directly above or below the origin). The parameter ' b ' determines the slope of the line, i.e., the change in Y for unit change in X. The symbol Y_c stands for the value Y computed from the relationship for a given X.

If the values of the constants ' a ' and ' b ' are obtained, the line is completely determined. But the question is how to obtain these values. The answer is provided by the method of least squares which states that the line should be drawn through the plotted points in such a manner that the sum of the squares of the vertical deviations of the actual Y values from the computed Y values is the least, or, in other words, in order to obtain a line which fits the points best $\sum (Y - Y_e)^2$ should be minimum*.

Such a line is known as the line of best fit.

$$\sum (Y - Y_c)^2$$

* should be minimum ,or

$$\sum (Y - a - bX)^2 \quad [Y_c = a + bX]$$

should be minimum

Let

Differentiating partially with respect to a and b,

$$\begin{aligned} \frac{\partial S}{\partial a} &= \sum (Y - a - bX)(-1) = 0 \\ \sum (Y - a - bX) &= 0 \\ \sum Y &= \sum a + b \sum X \dots\dots\dots (1) \\ \frac{\partial S}{\partial b} &= \sum (Y - a - bX)(-X) = 0 \\ \sum (Y - a - bX)(-X) &= 0 \\ \sum (Y - a - bX)(-X)X &= 0 \\ \sum XY &= a \sum X + b \sum X^2 \dots\dots\dots (2) \end{aligned}$$

With a little algebra and differential calculus, it can be shown that the following two equations, if solved simultaneously, will yield values of the parameters a and b such that the least squares requirement is fulfilled:

$$\begin{aligned} \sum Y &= Na + b \sum X \dots\dots\dots (i) \\ \sum XY &= a \sum X + b \sum X^2 \dots\dots\dots (ii) \end{aligned}$$

These equations are usually called the normal equations. In the equations $\sum X, \sum Y, \sum XY, \sum X^2$ indicate totals which are computed from the observed pairs of values of two variables X and Y to which the least squares estimating line is to be fitted and N is the total number of observed pairs of values.

Solving the equations

$$a = \frac{\sum X}{N} - b \frac{\sum Y}{N}$$

$$b_{yx} = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

This is called regression coefficient .

Regression coefficient of Y on X , $b_{YX} = r \frac{\sigma_Y}{\sigma_X}$

Regression Equation of X on Y

The regression equation of X on Y is expressed as follows:

$$X_c = a + bY$$

To determine the values of a and b the following two normal equations are to be solved simultaneously.

$$\begin{aligned} \sum X &= Na + b \sum Y \dots\dots\dots(i) \\ \sum XY &= a \sum Y + b \sum Y^2 \dots\dots\dots(ii) \end{aligned}$$

Solving the equations

$$a = \frac{\sum Y}{N} - b \frac{\sum X}{N}$$

$$b_{xy} = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum y^2) - (\sum y)^2}$$

This is called regression coefficient .

Regression coefficient of X on Y , $b_{xy} = r \frac{\sigma_X}{\sigma_Y}$

Also The regression of X on Y is different from the regression of Y on X. Thus, there are two equations of regression and the two regression lines are given as follows:

$$\text{Regression of Y on X: } Y - \bar{Y} = b_{yx}(X - \bar{X})$$

$$\text{Regression of X on Y: } X - \bar{X} = b_{xy}(Y - \bar{Y})$$

Where $\bar{X}$, $\bar{Y}$ are the means of X, Y respectively.

Example

You are provided with the following set of data:

Advertising Expenditure	Sales
(X) in Tk lakh	(Y) in Tk lakh
5	10
4	8
3	3
4	8
5	7

- (i) Find out coefficient of determination.
- (ii) Estimate the regression equation that best fits the data.
- (iii) Also estimate the sales when advertising expenditure is Tk 20 lakh.

Solution:

Calculation of Coefficient of Determination (r^2)

x	y	x^2	y^2	xy
5	10	25	100	50
4	8	16	64	32
3	3	9	9	9

4	8	16	64	32
5	7	25	49	35
$\Sigma x = 21$	$\Sigma y = 36$	$\Sigma x^2 = 91$	$\Sigma y^2 = 286$	$\Sigma xy = 158$

$$\begin{aligned}
 \text{(i) Here } r &= \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{n(\sum x^2) - (\sum x)^2} \cdot \sqrt{n(\sum y^2) - (\sum y)^2}} \\
 &= \frac{5(158) - (21)(36)}{\sqrt{5(91) - (21)^2} \cdot \sqrt{5(286) - (36)^2}} \\
 &= 0.78
 \end{aligned}$$

$\therefore$ The coefficient of determination $(r^2) = (r)^2$

$$= (0.78)^2$$

$$= 0.61 \quad [\text{Ans.}]$$

(ii) Here $\bar{x} = \frac{\sum x}{n} = \frac{21}{5} = 4.2$

$$\bar{y} = \frac{\sum y}{n} = \frac{36}{5} = 7.2$$

$$b_{yx} = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2} = \frac{5(158) - (21)(36)}{5(91) - (21)^2} = 2.43$$

So the regression equation that best fits the data is

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$\Rightarrow y - 7.2 = 2.43(x - 4.2)$$

$$\Rightarrow y - 7.2 = 2.43x - 10.21$$

$$\Rightarrow y = 2.43x - 3.01 \text{ [Ans.]}$$

(iii) If advertisement (x) = 20, then the estimated sales would be

$$y = 2.43x - 3.01 = 2.43(20) - 3.01 = 45.59 \text{ [Ans.]}$$

Example You are provided with the following set of data:

Advertising Expenditure (X)	1	2	3	4	5	6	7	8	9
Sales(Y)	9	8	10	12	11	13	14	16	15

i) Find correlation coefficient.

ii) Find regression coefficients Y on x and X on Y.

iii) Find regression equations Y on x and X on Y.

iv) Plot the regression equation (line) y on x on a graph paper .

Solution: Calculation of Correlation Coefficient and Regression Equations

x	y	x^2	y^2	xy
1	9	1	81	9
2	8	4	64	16
3	10	9	100	30
4	12	16	144	48
5	11	25	121	55
6	13	36	169	78
7	14	49	196	98
8	16	64	256	128
9	15	81	225	135
$\Sigma x = 45$	$\Sigma y = 108$	$\Sigma x^2 = 285$	$\Sigma y^2 = 1356$	$\Sigma xy = 597$

(i) The value of correlation coefficient is

$$\begin{aligned}
 r &= \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{n(\sum x^2) - (\sum x)^2} \cdot \sqrt{n(\sum y^2) - (\sum y)^2}} \\
 &= \frac{9(597) - (45)(108)}{\sqrt{9(285) - (45)^2} \cdot \sqrt{9(1356) - (108)^2}} \\
 &= 0.95 \text{ [Ans.]}
 \end{aligned}$$

ii) Regression equation y on x

$$Y = a + b x \text{ and}$$

X on y

$$X = a + by$$

Regression coefficients

$$b_{yx} = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2} = \frac{9(597) - (45)(108)}{9(285) - (45)^2} = 0.95$$

$$b_{xy} = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum y^2) - (\sum y)^2} = \frac{9(597) - (45)(108)}{9(1356) - (108)^2} = 0.95$$

iii) Here $\bar{x} = \frac{\sum x}{n} = \frac{45}{9} = 5$ $\bar{y} = \frac{\sum y}{n} = \frac{108}{9} = 12$

So the regression equation of y on x is

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$\Rightarrow y - 12 = 0.95(x - 5)$$

$$\Rightarrow y - 12 = 0.95x - 4.75$$

$$\Rightarrow y = 0.95x + 7.25$$

Regression equation $Y = 7.25 + 0.95x$

So the regression equation of x on y is

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$\Rightarrow x - 5 = 0.95(y - 12)$$

$$\Rightarrow x - 5 = 0.95y - 11.4$$

$$\Rightarrow x = 0.95y - 6.4$$

Regression equation $x = -6.4 + 0.95y$

x	y	x ²	y ²	xy	Y(cal)
1	9	1	81	9	8.20
2	8	4	64	16	9.15
3	10	9	100	30	10.10
4	12	16	144	48	11.05
5	11	25	121	55	12.00
6	13	36	169	78	12.95
7	14	49	196	98	13.90
8	16	64	256	128	14.85
9	15	81	225	135	15.80
$\Sigma x = 45$	$\Sigma y = 108$	$\Sigma x^2 = 285$	$\Sigma y^2 = 1356$	$\Sigma xy = 597$	

Regression equation $Y = 7.25 + 0.95x$

Example:

Obtain two regression equations from the following information about advertising expenditure and sales:

	Ad. Exp. (lakh)	Sales (lakh)
$\bar{X}$	10	90
σ	3	12
Correlation coefficient = + 0.80		

Solution: (a) Regression equation of X on Y is

$$\begin{aligned}
 X - \bar{X} &= r \frac{\sigma_x}{\sigma_y} (Y - \bar{Y}) \\
 \Rightarrow X - 10 &= 0.20 (Y - 90) \\
 \Rightarrow X - 10 &= 0.20 Y - 18 \\
 \Rightarrow X &= 0.20 Y - 8 \quad [\text{Ans}]
 \end{aligned}$$

where,

$$r \frac{\sigma_x}{\sigma_y} = .8 \frac{3}{12} = 0.20$$

(b) Regression equation of Y on X is

$$\begin{aligned}
 Y - \bar{Y} &= r \frac{\sigma_y}{\sigma_x} (X - \bar{X}) \\
 \Rightarrow Y - 90 &= 3.2 (X - 10) \\
 \Rightarrow Y - 90 &= 3.2 X - 32 \\
 \Rightarrow Y &= 3.2 X + 58 \quad [\text{Ans}]
 \end{aligned}$$

where,

$$r \frac{\sigma_y}{\sigma_x} = .8 \frac{12}{3} = 3.2$$

Example

Number of man-hours and the corresponding productivity (in units) are furnished below. Fit a simple linear regression equation $\hat{Y} = a + bx$ applying the method of least squares.

Man-hours	3.6	4.8	7.2	6.9	10.7	6.1	7.9	9.5	5.4
Productivity (in units)	9.3	10.2	11.5	12	18.6	13.2	10.8	22.7	12.7

Solution:

The simple linear regression equation to be fitted for the given data is

$$\hat{Y} = a + bx$$

Here, the estimates of a and b can be calculated using their least squares estimates

$$\begin{aligned} \hat{a} &= \bar{y} - \hat{b}\bar{x} \\ \text{i.e., } \hat{a} &= \frac{1}{n} \sum_{i=1}^n y_i - \hat{b} \frac{1}{n} \sum_{i=1}^n x_i \\ \hat{b} &= \frac{\frac{1}{n} \sum_{i=1}^n x_i y_i - (\bar{x} \times \bar{y})}{\frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2} \\ \text{or equivalently } \hat{b} &= \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \times \sum_{i=1}^n y_i \right)}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} \end{aligned}$$

From the given data, the following calculations are made with $n=9$

Man-hours x_i	Productivity y_i	x_i^2	$x_i y_i$
3.6	9.3	12.96	33.48
4.8	10.2	23.04	48.96
7.2	11.5	51.84	82.8
6.9	12	47.61	82.8
10.7	18.6	114.49	199.02
6.1	13.2	37.21	80.52
7.9	10.8	62.41	85.32
9.5	22.7	90.25	215.65
5.4	12.7	29.16	66.42
$\sum_{i=1}^9 x_i = 62.1$	$\sum_{i=1}^9 y_i = 121$	$\sum_{i=1}^9 x_i^2 = 468.97$	$\sum_{i=1}^9 x_i y_i = 894.97$

Substituting the column totals in the respective places in the of the estimates $\hat{a}$ and $\hat{b}$, their values can be calculated as follows:

$$\begin{aligned}
 \hat{b} &= \frac{(9 \times 894.97) - (62.1 \times 121)}{(9 \times 468.97) - (62.1)^2} \\
 &= \frac{8054.73 - 7514}{4220.73 - 3856.41} \\
 &= \frac{540.73}{364.32}
 \end{aligned}$$

Thus, $\hat{b} = 1.48$.

Now $\hat{a}$ can be calculated using $\hat{b}$ as

$$\begin{aligned}
 \hat{a} &= 121/9 - (1.48 \times 62.1/9) \\
 &= 13.40 - 10.21
 \end{aligned}$$

Hence, $\hat{a} = 3.19$

Therefore, the required simple linear regression equation fitted to the given data is

$$\hat{Y} = 3.19 + 1.48x$$

Properties of Regression coefficients

1. The coefficient of correlation is the geometric mean of the two regression coefficients. Symbolically:

$$r = \pm \sqrt{b_{YX} b_{XY}}$$

Proof: Regression coefficient of Y on X , $b_{YX} = r \frac{\sigma_Y}{\sigma_X}$ (i)

Regression coefficient of X on Y , $b_{XY} = r \frac{\sigma_X}{\sigma_Y}$ (ii)

Multiplied equation (i) & (ii), we get,

$$b_{YX} \times b_{XY} = r \frac{\sigma_Y}{\sigma_X} \times r \frac{\sigma_X}{\sigma_Y}$$

$$r^2 = b_{YX} \times b_{XY}$$

$$r = \pm \sqrt{b_{YX} * b_{XY}} \text{ (Proved)}$$

2. If one of the regression coefficients is greater than unity the other must be less than unity.

3. Both the regression coefficients will have the same sign, i.e. they will be either positive or negative.

4. The coefficient of correlation will have the same sign as that of regression coefficients, i.e., if regression coefficients have positive sign, then r is positive. If regressions coefficient have negative sign then r is negative.

For Example: If $b_{YX} = -0.1$ & $b_{XY} = -10$ Then $r = \pm \sqrt{-0.1 \times -10} = -1$

5. The average value of the two regression coefficients would be greater than the value of the correlation coefficient.

6. Regression coefficients are independent of change of origin but not of scale.

6.18 Distinguish between Correlation & regression

Correlation	Regression
1. The degree of relationship between two or variables is called correlation.	1. Regression means stepping back or returning to the average value, i.e. it expresses average relationship between two or more variables.
2. Correlation need not imply cause & effect relationship.	2. Regression analysis clearly indicates the cause & effect relationship. The variable(s) constituting cause(s) is taken as independent variable(s) & the variable constituting the effect is taken as dependent variable.
3. It may be a case of nonsense correlation.	3. There is nothing like nonsense regression.
4. Correlation coefficient between X & Y & Correlation coefficient between Y & X are same i.e. $r_{XY} = r_{YX}$	4. But Regression coefficient of X on Y and Regression coefficient of Y on X are not same i.e. $b_{XY} \neq b_{YX}$.
5. Correlation Coefficient lies between -1 & +1.	5. Regression coefficient lies between $-\infty$ to $+\infty$.
6. Correlation analysis studies only linear relationship.	6. Regression analysis studies linear as well as non linear relationship.

Questions

1. Prove that $-1 \leq r \leq +1$.
2. Discuss the concepts of correlation and regression.
3. Distinguish between correlation and regression.
4. Distinguish between positive and negative correlation with example.
5. What is scatter diagram? How does it help in studying the correlation between two variables?
6. What is correlation? Discuss different types of correlation..
7. What is the significance of measuring correlation?
8. Explain the following values of 'r' with diagram:
(i) $r = 0$; (ii) $r = -1$; (iii) $r = +1$.
9. Interpret: $r = -1$; $r = 0$, $r = 0.785$; $r = -.05$.
10. State the relationship between coefficient of correlation and regression coefficient.
11. What is regression coefficient? How does it differ from a correlation coefficient?

12. State some of the important properties of regression coefficient.
13. What is linear regression? When is it used? How estimate regression parameters with Least square method.
14. Define the Regression coefficient. Write down its properties.

Exercises

Exercise 1

Given the following pairs of values:

Capital employed (Tk in crore)	1	2	3	4	5	7	8	9	11	12
Profits (Tk in lakh)	3	5	4	7	9	8	10	11	12	14

- (a) Make a scatter diagram.
- (b) Do you think that there is any correlation between profits and capital employed? Is it positive? Is it high or low?

Exercise 2

Find the correlation coefficient between the sales and expenses from the data given below:

Firm	1	2	3	4	5	6	7	8	9	10
Sales (Tk in lakh)	50	50	55	60	65	65	65	60	60	50
Expenses (Tk in lakh)	11	13	14	16	16	15	15	14	13	13

[Answer: $r = 0.787$]

Exercise 3

The following data relate to the age of 10 employees and the number of days which they reported sick in a month:

Age	20	30	32	35	40	46	52	55	58	62
Sick days	11	12	10	13	14	16	15	17	18	19

Calculate Karl Pearson's coefficient of correlation and interpret its value.

[Answer: $r = 0.939$]

Exercise 4

Calculate the coefficient of correlation from the following data:

Adv. Exp. (Tk in lakh) X:	10	12	15	23	20
Sales (Tk in lakh) Y:	14	17	23	25	21

[Answer: $r = 0.865$]

Exercise 5

Two managers are asked to rank a group of employees in order of potential for eventually becoming top managers. The rankings are as follow:

Employees	Ranking by Manager I	Ranking by Manager II
A	10	9
B	2	4
C	1	2

D	4	3
E	3	1
F	6	5
G	5	6
H	8	8
I	7	7
J	9	10

Compute the coefficient of rank correlation and comment on the value.

[Answer: $R = 0.915$]

Exercise 6

Ten competitors in a beauty contest are ranked by three Judges in the following order:

1st Judge	1	6	5	1	3	2	4	9	7	8
			0							
2nd Judge	3	5	8	4	7	10	2	1	6	9
3rd Judge	6	4	9	8	1	2	3	10	5	7

Use the rank correlation coefficient to determine which pair of Judges has the nearest approach to common tastes in beauty?

[Answer: $R(1 \text{ \& } 2) = -0.212$, $R(2 \text{ \& } 3) = -0.297$, $R(1 \text{ \& } 3) = 0.636$.]

Exercise 7

The coefficient of rank correlation between debenture prices and share prices is found 0.143 and the sum of squares of the difference is 48. Find out the value of N.

[Answer: $N = 7$]

Exercise 8

Calculate the rank correlation coefficient for the following data of marks of 2 tests given to candidates for a clerical job.

Preliminary test	92	89	8	86	83	77	71	6	53	50
			7					3		
Final test	86	83	9	77	68	85	52	8	37	57

			1				2		
--	--	--	---	--	--	--	---	--	--

[Answer: R = 0.733.]

Exercise 9

An examination of eight applicants for a clerical post was taken by a firm. From the marks obtained by the applicants in the Accountancy and Statistics papers, compute rank coefficient of correlation.

Applicant	A	B	C	D	E	F	G	H
Marks in Accountancy	15	20	28	12	40	60	20	80
Marks in Statistics	40	30	50	30	20	10	30	60

[Answer: R = 0]

Exercise10

You are given the following information about advertising expenditure and sales:

	Adv. Exp. (X)	Sales (Y)
	(Tk in lakh)	(Tk in lakh)
Mean	10	90
S.D.	3	12

Correlation coefficient = 0.8

- Calculate the two regression equations.
- Find the likely sales when advertisement budget is Tk 15 lakh.

- (iii) What should be the advertisement budget if the company wants to attain sales target of Tk 120 lakh?

[Answers: (i) $Y = 3.2X + 58$, $X = 0.2Y - 8$; (ii) $Y = 106$; (iii) $X = 16$.]

Exercise 11

Calculate the regression equations of X on Y and Y on X from the following data:

X	1	2	3	4	5
Y	2	5	3	8	7

[Answers: (i) $X = 0.5 + 0.5Y$, (ii) $Y = 1.10 + 1.3X$]

Exercise 12

In the following table are recorded data showing the test scores made by salesmen on an intelligence test and their weekly sales:

Salesmen	1	2	3	4	5	6	7	8	9	10
Test score	40	70	50	60	80	50	90	40	60	60
Sales ('000 Tk)	2.5	6.0	5.0	5.0	4.0	2.5	5.5	3.0	4.5	3.0

Calculate the regression equation of sales on test scores and estimate the probable weekly sales volume if a salesman makes a score of 100.

[Answers: (i) $Y = 0.76 + 0.054X$, (ii) $Y = 6.16$]

Exercise 12

A company wants to assess the impact of R & D expenditure on its annual profit. The following table presents the information for the last eight years:

Years	2013	2012	2011	2010	2009	2008	2007	2006
R & D expenditure (Tk '000)	9	7	5	10	4	5	3	2
Annual Profit (Tk '000)	45	42	41	60	30	34	25	20

Estimate the regression equation and predict the annual profit for 2019 for an allocated sum of Tk 100000 as R & D expenditure.

[Answers: (i) $Y = 13.129 + 4.266X$, (ii) $Y = 439.729$ thousand.]

Exercise 13

From the following data obtain two regression equations:

Sales	91	9	108	121	6	124	51	7	111	57
		7			7			3		
Purchase	71	7	69	97	7	91	39	6	80	47
		5			0			1		

[Answers: (i) $X = -5.2 + 1.36Y$, (ii) $Y = 14.83 + 0.613X$]